Cover letter

Mathematics of Computation 25 October 2026

Unsent submission correspondence; prepared for author review.

Dear Editors,

Please consider the manuscript “Lehmer’s totient problem with fewer than sixteen prime factors” by Deep Bhattacharjee.

The research revision establishes the stated sixteen-prime-factor lower bound and companion-equation results through written finite reductions and computations. It adds a one-candidate reduction for each prime prefix and independently checked primality certificates. Lehmer’s full totient conjecture remains open.

The finite reductions, reproducible computational number theory, independently checked searches and primality certificates make a computational-number-theory journal the clearest fit.

The prior public preprint is `https://hexagonmath.org/2610.00454v1`. The research manuscript/source archive is `https://doi.org/10.5281/zenodo.23617604`. Editable sources and verification records are supplied with the manuscript.

The repository contains partial Lean results for statements not depending on the exhaustive search. The full searches and the full Lehmer conjecture are not claimed to be Lean-formalised.

The manuscript and the accompanying declaration disclose the substantive use of Claude and OpenAI tools. The submitting author has declared that all parts have been checked and understood by a human. The named human authors retain responsibility for the content.

Sincerely,

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